



LLM capacity & dimensioning studio: model, precision, workload and hardware in; memory fit, latency, throughput and resilient topology out.

GPUScale.net · Untitled scenario · Kimi K3 2.8T-A50B (MoE) · weights FP8 / KV FP8 · seq 16K (+7.8K reasoning) · 377 concurrent, batch 4/replica · 8× worker (8 GPU) B300 288GB · TP8 · Active/Active · two live sites (2N) → 16 workers procured · 7/18/2026

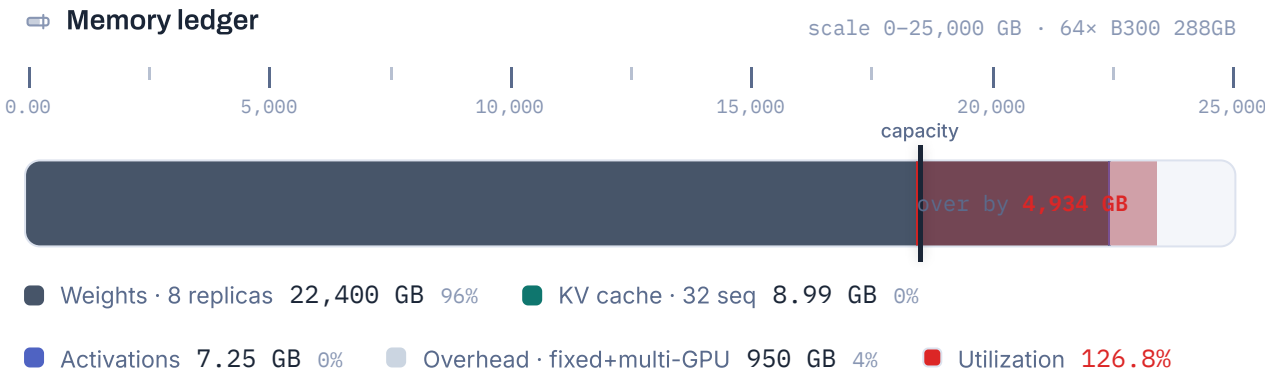
●

Exceeds VRAM

23,366 GB required

18,432 GB across 64× GPU

Needs 23,366 GB but only 18,432 GB available: quantize further, cut context or batch, or add workers.



⚡ Time to first token

182_{ms}

prefill 16K tok on TP8 · MFU 0.5

🔄 Per-user speed

692_{tok/s}

ITL 1.45 ms/tok · batch/replica 4.00

📊 Aggregate throughput

22,133_{tok/s}

32 admitted × per-user · 345 queued

🕒 Mean user latency

12.9_s

P95 ≈ 16.8 s · generates 8,800 tok

🕒 SLO compliance

targets from "Internal GPT / Copilot"



TTFT

182 ms · ≤ 1,000 ms



Per-user TPS

692 tok/s · ≥ 25.0



P95 latency

16.8 s · ≤ 30.0 s

🔗 Deployment topology · workers & GPU utilization

A/A · 8 replicas × TP8 · 127%/GPU

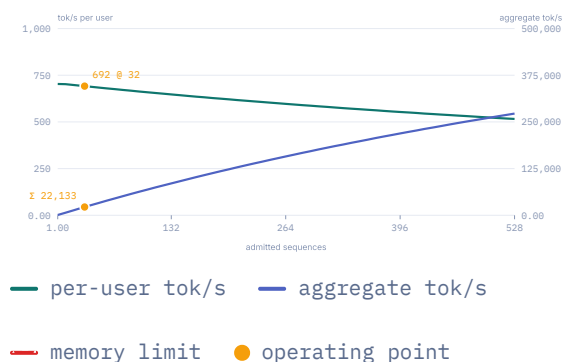


■ active worker · GPU bars show memory fill (127% of 288 GB each)

Load: **N = 8 workers · 64 GPUs** (8/worker): performance and fit are computed on these. Procured for Active/Active · two live sites (2N): **16 workers · 128 GPUs · ≈ 154 kW** GPU TDP. Two active sites share traffic behind global load balancing. Each site alone can carry the full load, so losing a site degrades nothing; in normal operation each runs at roughly half load.

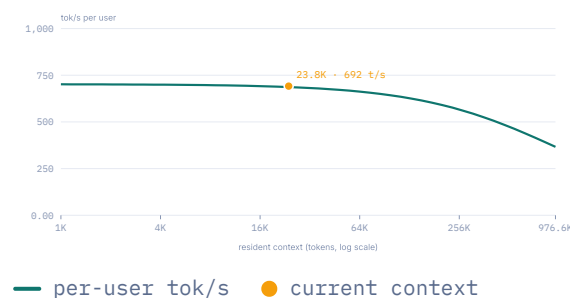
↗️ Throughput vs admitted sequences

batch-aware



~ Per-user speed vs context length

at current batch



🕒 Request latency anatomy

mean 12.9 s · p95 16.8 s



■ TTFT 182 ms ■ Overhead 30.0 ms ■ Reasoning 11.6 s ■ Visible output 1.16 s

💡 Reading of this configuration

- ▲ 345 of 377 calls queue at peak (only 32 admitted). Memory allows up to 0 per replica: add workers or trim context to admit more.
- ▶ Marginal cost of one more admitted call: 0.28 GB of KV. Speculative decoding (EAGLE-3 / MTP) would add 1.5–3× on top of the decode figures.

📝 Recommendations

primary bottleneck: VRAM capacity

Configuration does not fit

Needs 23,366 GB against 18,432 GB of serving VRAM, over by 4,934 GB. Options: add ≈3 workers, trim resident context or batch.

Calls queue at peak

345 of 377 concurrent calls wait; only 32 are admitted. Admitting everyone at batch 4 needs ≈95 workers (currently 8).

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Capacity & dimensioning studio for self-hosted LLM deployments. All figures are peak estimates; production typically achieves 70–90%. Validate with vLLM bench / GenAI-Perf before commitments.

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Version Studio 4.8.0

Engine v23 · library v24 (2026-07-18)

Scripting API at `window.GPUScale`

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